

K-Field Lab-Scale Device Proposals

Force-optimized circularly polarized elastic oscillators, balanced high-overtone acoustic arrays, metrology, controls, and terahertz material extensions

Proposed three-module HBAR force package - engineering dimensions



Reference engineering layout for the recommended three-module force-optimized laboratory device.

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This paper is a standalone archival and educational reference. It defines the Cell3 geometry, source field state, circularly polarized elastic-oscillator force law, force-optimization criterion, recommended laboratory architecture, expected output envelope, calibration procedures, force-balance design, decisive controls, and the progression from megahertz acoustic devices to terahertz coherent ionic motion. All numerical tables and plotted quantities are generated from the supplied machine-readable records or from engineering parameters explicitly introduced and labeled in this paper.

Abstract

The force law developed for circularly polarized elastic motion makes the active-mass-frequency product the primary ideal device figure of merit. The apparent instruction to use minimal mass is therefore refined: inactive package mass should be minimized, while coherently participating resonator mass should be preserved. The strongest laboratory-scale implementation is a three-module dual-polarization high-overtone bulk acoustic resonator array. Each module generates a circular material trajectory from two orthogonal shear modes driven in quadrature. Carrier phases of 0, 120, and 240 deg cancel ordinary first-harmonic reaction and the predicted second-harmonic ripple, while identical local chirality preserves and adds the phase-independent lateral mean. A preferred design range of 30 to 100 mg active mass at 50 to 100 MHz gives an ideal output envelope from 0.135 to 0.899 N before multiplication by measured circularity, orientation, and transfer factor H . The apparatus is designed to measure H as a function of amplitude, frequency, wave vector, material, temperature, and boundary state. SrTiO₃ driven at 3 THz and LiNbO₃ chiral phonon modes provide later-stage high-frequency scale tests after the megahertz coherence surface is established.

Primary conclusion

The recommended first device is a three-module, dual-polarization high-overtone bulk acoustic resonator array. It maximizes the product of coherent active mass and frequency, minimizes nonparticipating package mass, creates reversible local mechanical chirality, and uses temporal and spatial symmetry to reject ordinary reaction forces and torques.

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1. Scope, Design Objective, and Status Conventions

The objective is to identify a laboratory-scale device that maximizes the calculated lateral force while using high frequency, low inactive mass, and mechanically quiet operation. The work begins from the supplied circularly polarized elastic-oscillator reduction and the supplied material ledger for coherently controlled angular momentum density. The central engineering task is to preserve the local closed displacement loop and its handedness while removing the bearings, shafts, rotor imbalance, and continuous rigid-body circulation of a conventional rotor.

Five status classes are maintained throughout. Source-exact identities are carried directly from the supplied records. Derived identities follow algebraically from those source values. Engineering continuations introduce explicit dimensions, operating ranges, or implementation choices. Established transduction methods provide conventional means of exciting and measuring elastic modes. The unresolved physical transfer is represented by $H(\rho, \omega, k, \text{material, temperature, boundary})$, which must be mapped experimentally rather than treated as an arbitrary coefficient.

The calculated force outputs in this report are evaluations of the supplied force law. They are not substituted for measurements. This distinction makes the experimental program stronger: every ideal output has a defined normalization, and measured departures become direct information about coherent participation and localization.

2. Cell3 Coordinate Geometry and Dimensional Foundation

The Cell3 basis B maps integer address coordinates into the ICRS physical frame. The centered registry uses addresses $L = (i,j,k)$ with each coordinate in $\{-1,0,1\}$. A physical centered node is $p_L = B L / 2$. The registry therefore contains 27 points: eight corners, twelve edge centers, six face centers, and one body center. Antipodal addresses L and $-L$ map to opposite physical positions.

$$p_L = (1/2) B L \text{ and } x_L = B L$$

The primitive edge lengths are 0.334179679073 nm, 0.586410890415 nm, and 0.792319765038 nm. The included angles are 83.933483012 deg, 80.448130018 deg, and 50.363231814 deg. The cell volume is 0.117914891917 nm³. Because B is skew, equal address-channel amplitudes do not produce equal orthogonal physical displacements. Any physical circular orbit must be synthesized through B or through a directly measured actuator transfer matrix.

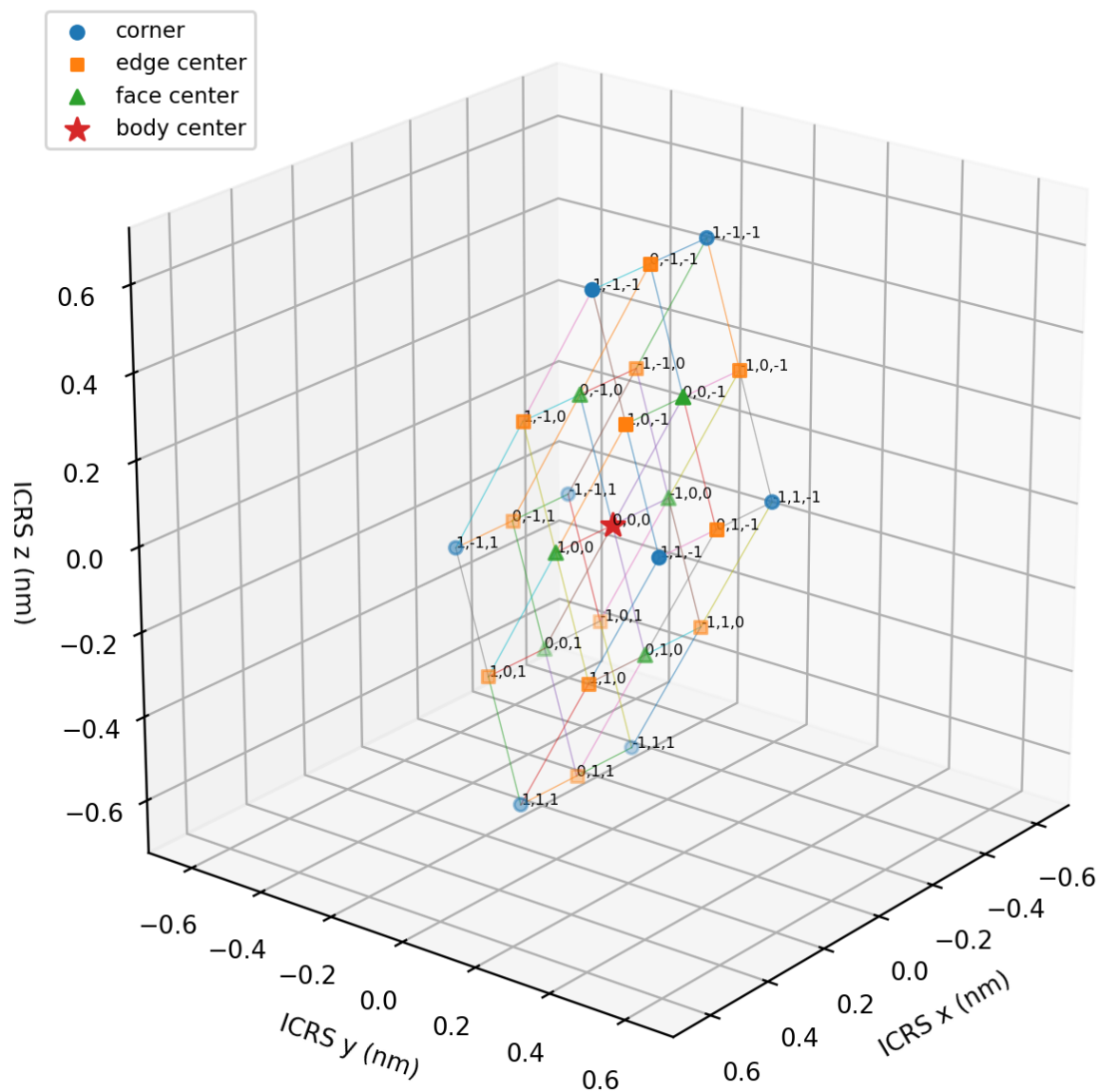
Cell3 centered 27-node registry: $p_L = B L / 2$ 

Figure 1. Dimensionally correct ICRS Cell3 base cell with all 27 centered nodes and address-neighbor connections.

Orthographic Cell3 views with every centered address

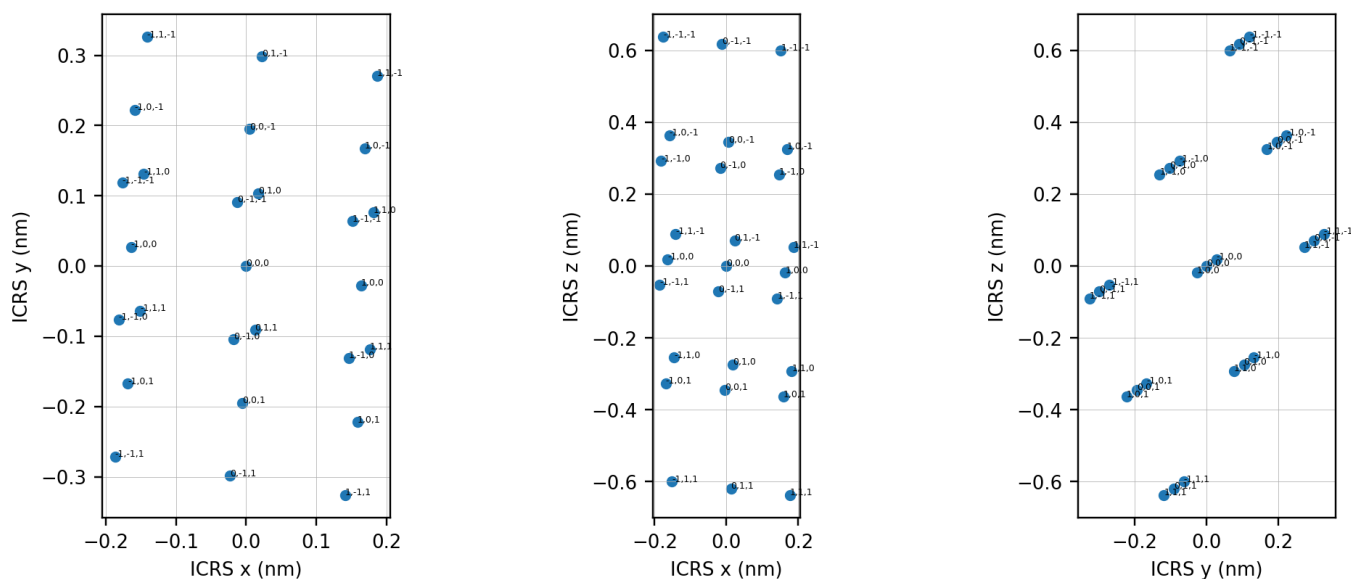


Figure 2. Orthographic projections of the complete 27-node Cell3 registry.

Direct metric, reciprocal metric, and response tensor

The direct metric $M = B^T B$ measures squared physical lengths of address vectors. The reciprocal metric $R = M^{-1}$ measures reciprocal-gradient lengths. The supplied address coupling W and physical response tensor G_{phase} obey the exact congruence relation below. The factor four follows from the centered half-step coordinate convention.

$$G_{\text{phase}} = 4 B^A (-T) W B^A (-1)$$

Quantity	Numerical value
B row 1 (nm)	0.32750124207, 0.035330775844, -0.010129317627
B row 2 (nm)	-0.054897007886, 0.207319850441, -0.389787702364
B row 3 (nm)	-0.037487502896, -0.547401085392, -0.689734408363
G row 1	2.9764106378, 1.2431507795, -2.0803600354
G row 2	1.2431507795, 2.5922286635, -2.3178839937
G row 3	-2.0803600354, -2.3178839937, 6.7735709133
Cross-source metric equivalence maximum error	1.140232e-11

The second source uses a canonical upper-triangular embedding of the same metric geometry. Its Gram matrix agrees with the ICRS embedding to the numerical error reported in the table. This provides a dimensionally equivalent frame for the seven-sheet representation while preserving the same edge lengths, angles, face areas, and cell volume.

Three coordinate-plane families

Cell3 a-channel plane family: bc faces; spacing 0.164773 nm

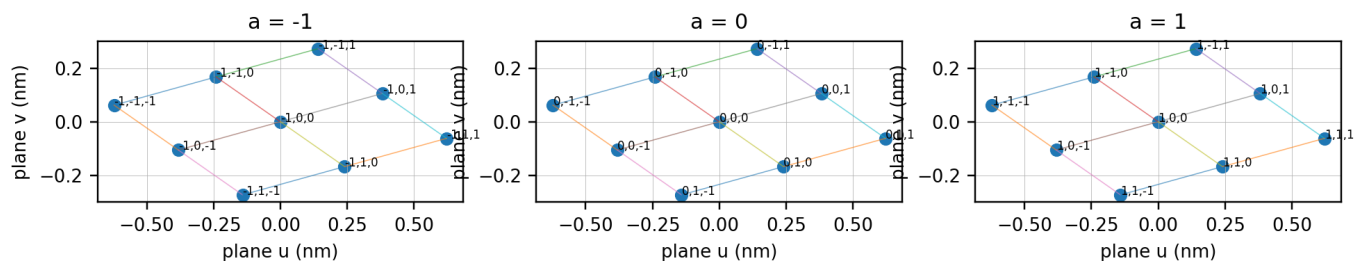


Figure 3. The a-channel planes. Primitive boundary spacing 0.329546529709 nm; centered adjacent-plane spacing 0.164773264855 nm; opposite face area 0.357809539129 nm².

Cell3 b-channel plane family: ac faces; spacing 0.225799 nm

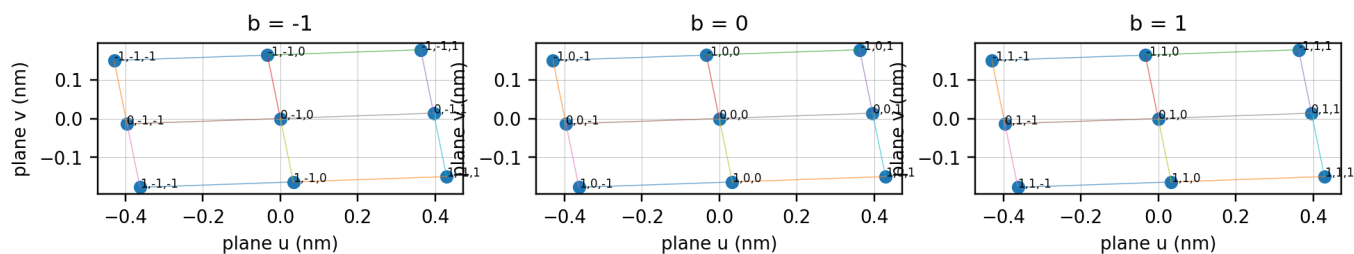


Figure 4. The b-channel planes. Primitive boundary spacing 0.451597379724 nm; centered adjacent-plane spacing 0.225798689862 nm; opposite face area 0.261106235800 nm².

Cell3 c-channel plane family: ab faces; spacing 0.302549 nm

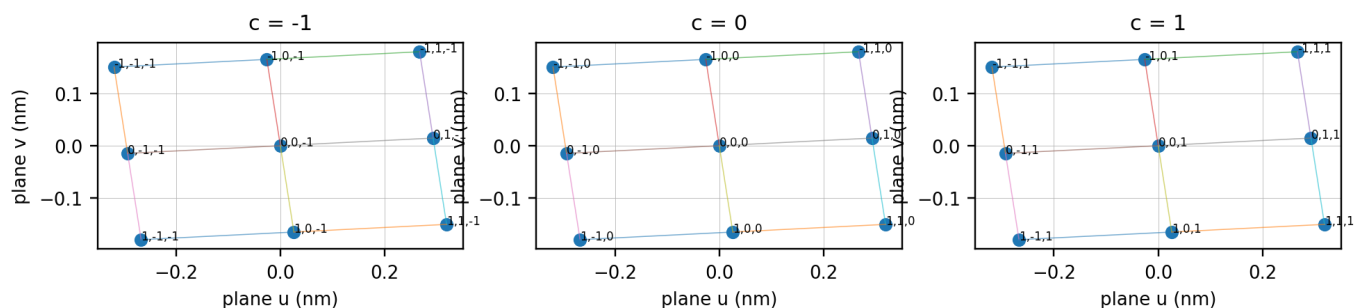


Figure 5. The c-channel planes. Primitive boundary spacing 0.605097741515 nm; centered adjacent-plane spacing 0.302548870757 nm; opposite face area 0.194869165470 nm².

Seven-sheet lambda family

The canonical metric-equivalent embedding also organizes the 27 nodes by $\lambda = i + j + k$. The sheet equation is $g_{111} \cdot x = \lambda/2$. There are seven sheets, $\lambda = -3$ through $+3$, with populations 1, 3, 6, 7, 6, 3, and 1. The adjacent sheet spacing is 0.151272688038 nm.

Seven-sheet g_{111} family: $g_{111} \cdot x = \lambda / 2$

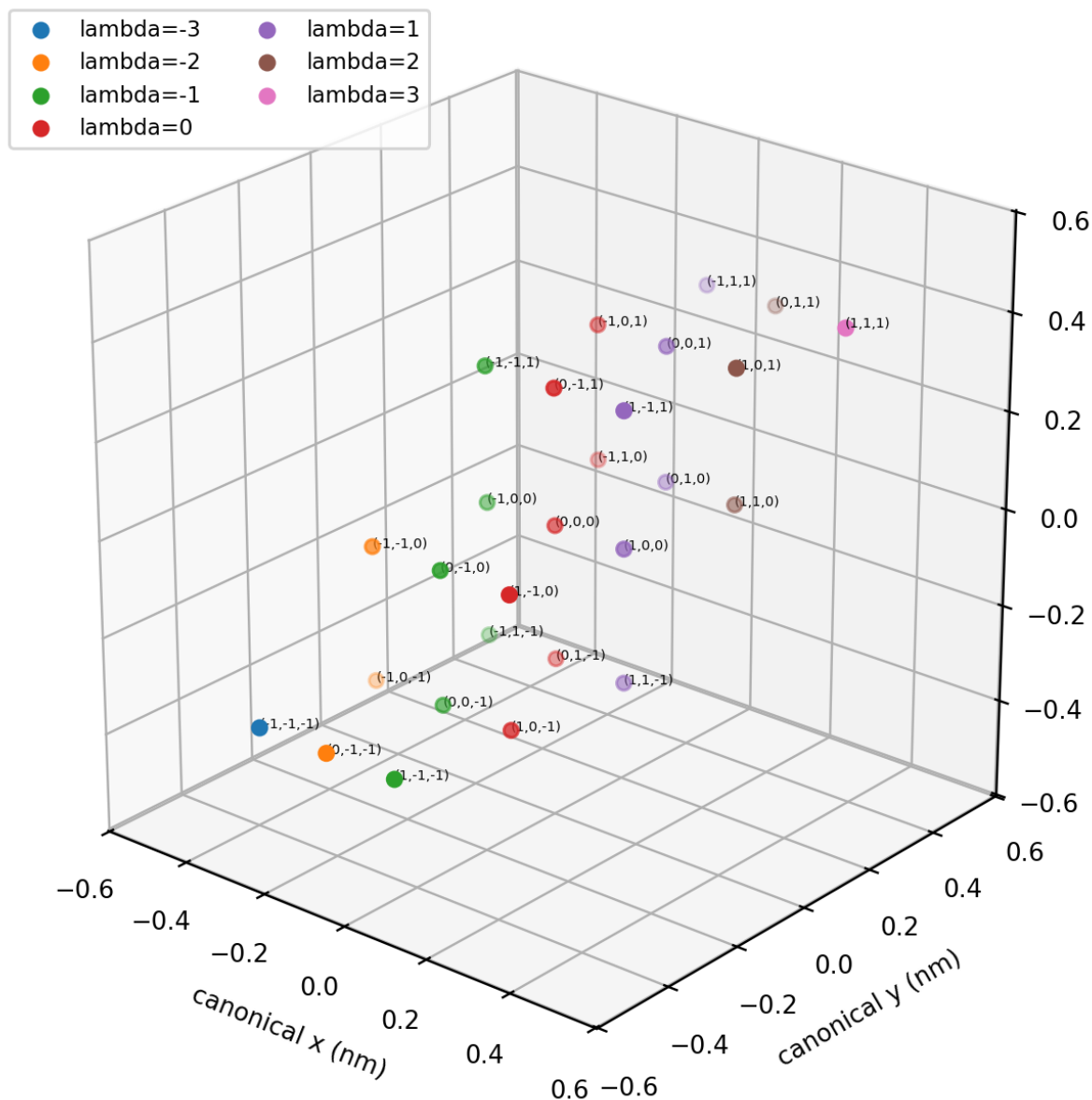


Figure 6. Complete seven-sheet lambda family with all nodes labeled.

3. Source Field State and Exact Circular-Oscillator Law

The Earth traversal state is represented by V_E in the ICRS frame. The physical response vector is $g = G_{\text{phase}} V_E$. The source values are V_E speed 369812.502000 m/s and $|g| = 930825.217270$ m/s. The maximum-output oscillator normal is perpendicular to g .

Quantity	ICRS components or value
V_E (m/s)	-3.590031990658e+05, 7.667988603400e+04, -4.470106010069e+04

Quantity	ICRS components or value
$g = G_{\text{phase}} V_E$ (m/s)	-8.802219816351e+05, -1.439114365616e+05, 2.663350269486e+05
$ g $ (m/s)	9.308252172704e+05
κ_F	3.074050899702035e-11
n_{opt}	-0.315865036407, 0.646162683024, -0.694768354089
lateral direction	0.077469585422, 0.747375863362, 0.65986951907
n_{opt} RA, Dec	116.050867377 deg, -44.008760899 deg
lateral RA, Dec	84.082114314 deg, 41.289922334 deg

Maximum-output oscillator plane and predicted lateral direction

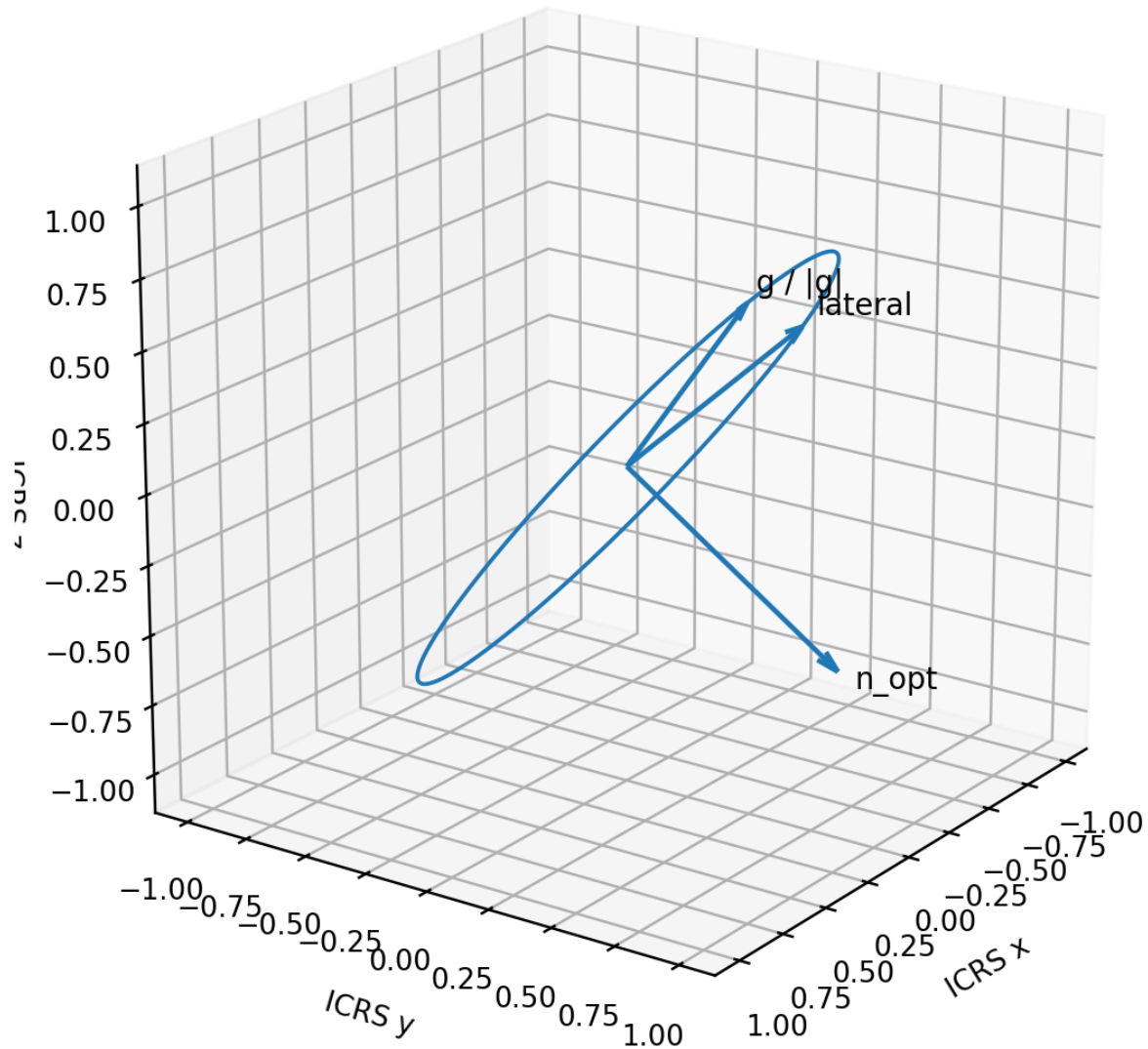


Figure 7. Maximum-output displacement plane, plane normal, and lateral direction.

For a circular material trajectory of handedness σ , frequency f , and loop normal n , the source-exact cycle velocity increment and mean acceleration are:

$$\Delta v_K = \pi \kappa_F \sigma (n \times g)$$

$$\langle a_K \rangle = \pi \kappa_F f \sigma (n \times g)$$

At maximum orientation, $|n \times g| = |g|$. The acceleration coefficient is therefore $\pi \kappa_F |g| = 8.989366088879e-5 \text{ m/s}^2$ per hertz. A complete device force is obtained by integrating over participating mass and multiplying by mechanical circularity, coherence, and orientation.

$$F_K = K_0 M_{\text{active}} f C_{\text{pol}} H \sin(\beta)$$

$$K_0 = 8.989366088879e-5 \text{ N} / (\text{kg Hz})$$

The circular trajectory radius cancels in the exact local differential result. One factor of radius enters the differential path length and the inverse radius enters the kernel. Amplitude nevertheless remains experimentally decisive because the source does not establish the continuation to arbitrarily small displacement. Any small-radius or high-frequency transition is collected in H and must be located by sweep measurements.

4. Mechanical Circularity and Minimum Sufficient Motion

Rigid-body rotation is not the minimum kinematic requirement. The minimum direct state contains two independent displacement modes. Let the orthogonal coordinates be $x_1 = A \cos(\omega t)$ and $x_2 = B \cos(\omega t - \phi)$. Their trajectory is an ellipse with signed area proportional to $A B \sin(\phi)$. Equal amplitudes and a relative phase of $+90$ or -90 deg produce a circle and maximize normalized oriented area.

$$C_{\text{pol}} = 2 A B \sin(\phi) / (A^2 + B^2)$$

C_{pol} equals zero for linear polarization and ± 1 for ideal circular polarization. Reversing ϕ from $+90$ to -90 deg reverses chirality and therefore the complete force vector while preserving nominal carrier frequency and RMS displacement. A one-axis reciprocal oscillator retraces its path and produces zero first-order cycle mean under the constant- κ source kernel.

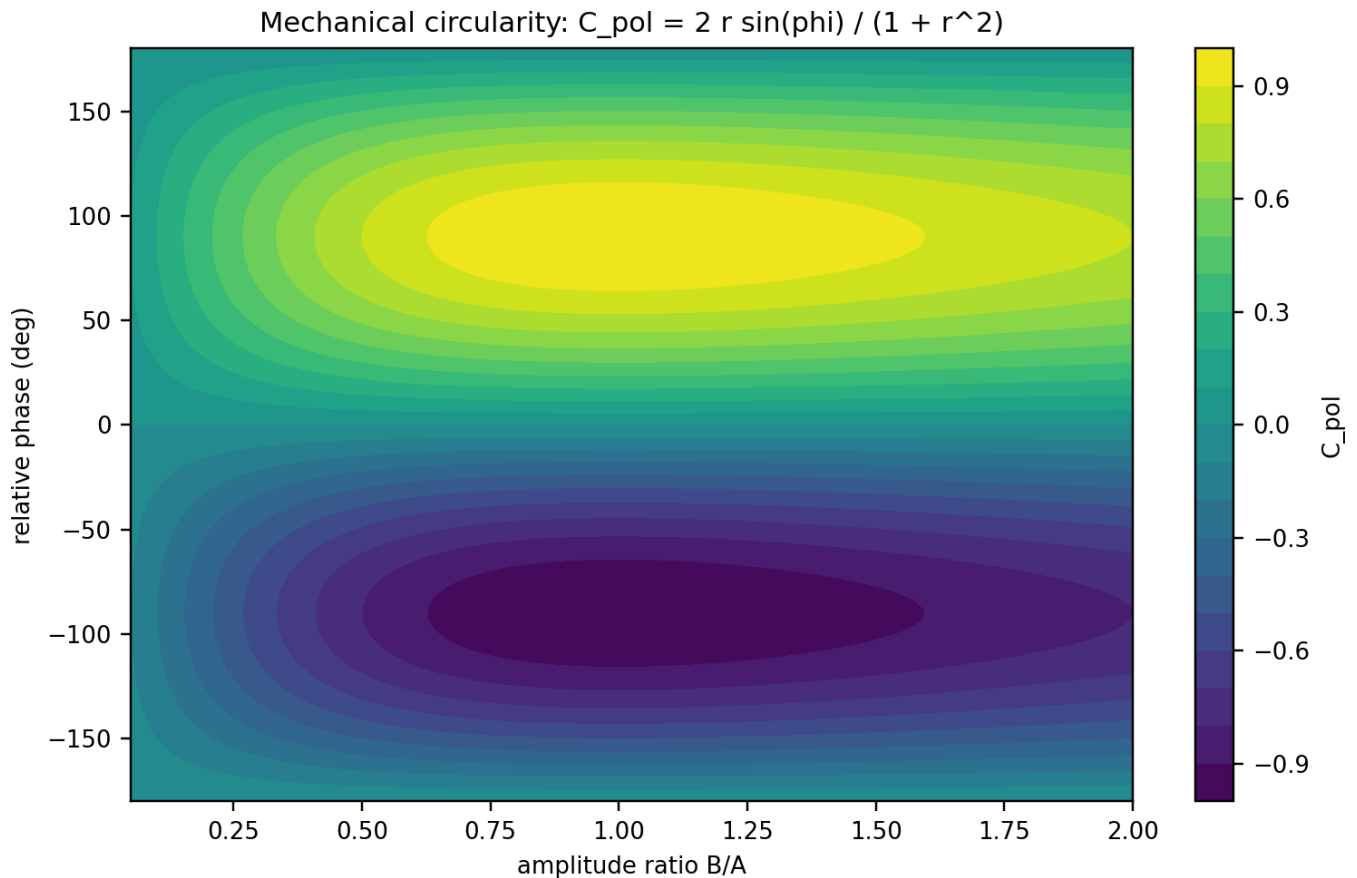


Figure 8. Mechanical circularity as a function of amplitude ratio and relative phase.

The target $C_{pol} \geq 0.95$ is a measured mechanical requirement, not an electrical assumption. Mode splitting, unequal damping, actuator cross-coupling, and temperature drift can cause voltage quadrature to differ from displacement quadrature. Each module therefore requires a complex modal estimator and closed-loop amplitude-phase control.

5. Force Optimization Criterion

The phrase minimal mass must be applied to the correct mass category. Active resonating mass produces the calculated force. Nonparticipating mounts, housings, electrodes, cabling, and instrumentation add weight without contributing. The force-per-total-mass objective is:

$$F_K / M_{total} = K_0 f \eta_m C_{pol} H \sin(\beta)$$

$$\eta_m = M_{active} / M_{total}$$

The optimized device maximizes frequency, active-mass fraction, circularity, coherence, and orientation simultaneously. Frequency alone is not sufficient. A microscopic resonator can lose its apparent advantage if participating mass falls faster than frequency rises. A macroscopic high-overtone substrate is therefore attractive because it can combine a high carrier frequency with milligram-scale coherent mass.

The ideal active-mass-frequency requirement for one newton is 11124.254926 kg Hz. Equivalent ideal combinations include 1 kg at 11.124 kHz, 11.124 g at 1 MHz, 11.124 mg at 1 GHz, and 11.124 micrograms at 1 THz. Physical equivalence across these cases depends on H and on the actual coherently active mass.

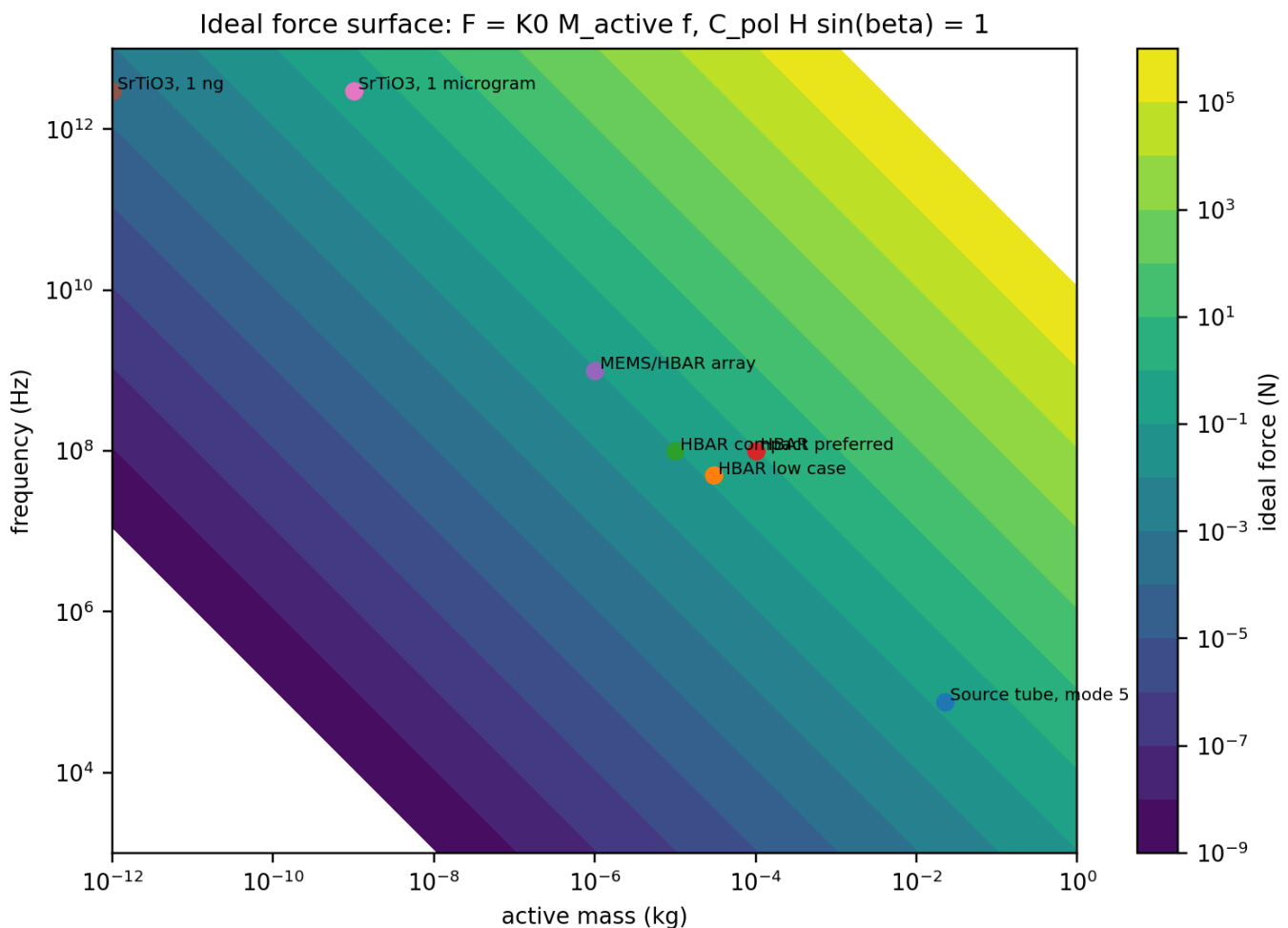


Figure 9. Ideal force surface and representative architecture points.

6. Architecture Comparison and Selection

Architecture	Primary advantage	Primary limitation	Recommended role
Whirling steel tube	Direct continuity with source tube; easy modal imaging	Frequency limited to the kilohertz range; attachments split modes	Bridge experiment and calibration
Radial-torsional annulus	Parallel local loop normals; monolithic balance	Requires matched radial and circumferential modes	Second macroscopic platform
Dual-shear HBAR	High frequency with substantial substrate mass	Active mass and mode circularity must be measured	Preferred force-optimized laboratory device
MEMS/NEMS array	Electronic phase coding and high frequency	Very small active mass; RF feedthrough	Scale and Fourier-array test
SrTiO ₃ coherent ionic orbit	3 THz circular soft mode	Picometre displacement and optical/THz backgrounds	Terahertz scale extension
LiNbO ₃ chiral phonons	6.77 to 25.39 THz reversible chiral modes	Coherent active mass and force readout difficult	Higher-frequency material cross-check

The strongest first implementation is the dual-shear HBAR array because it maximizes the product $M_{\text{active}} f$ without requiring a continuously rotating body. Alternating acoustic lobes reverse displacement and velocity together. Under the source parity rule, the local chiral term is unchanged by this simultaneous reversal, while conventional linear momentum can cancel strongly.

The source tube remains valuable as a bridge. Its first five calculated free-free bending frequencies are 5.700, 15.713, 30.803, 50.919, and 76.064 kHz for the stated steel geometry and an explicit 200 GPa modulus. The ideal full-mass force at the fifth mode is 154.66 mN. The HBAR proposal moves to tens or hundreds of megahertz while retaining tens of milligrams of active mass.

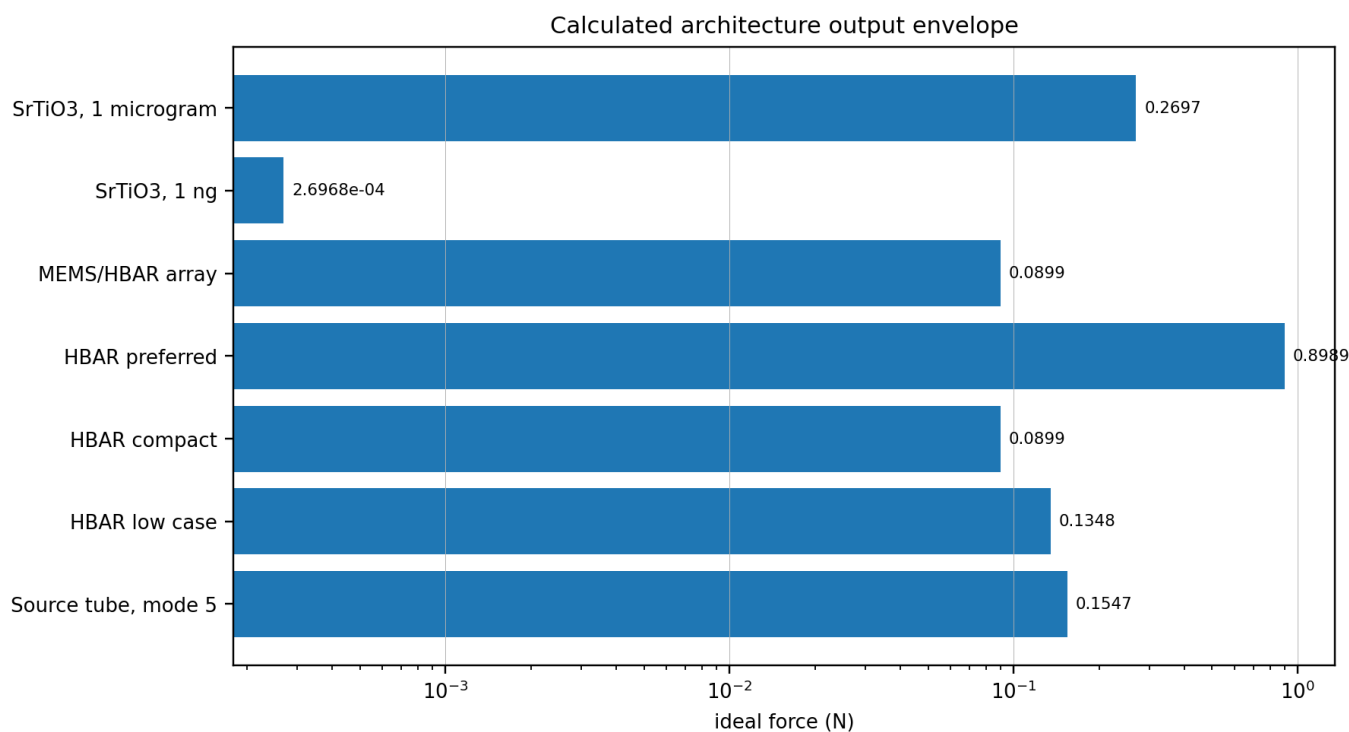


Figure 10. Calculated architecture output envelope.

7. Recommended Laboratory Device: Three-Module Dual-Shear HBAR Array

The recommended device contains three identical bulk-acoustic resonator modules positioned on an equilateral triangle and coupled to a common central flexure. Each resonator supports two orthogonal shear modes at the same selected overtone. The two modes are controlled to equal measured amplitude and ± 90 deg measured phase. All modules use the same chirality and loop normal. Their carrier starting phases are 0, 120, and 240 deg.

Parameter	Initial target
Total coherently active mass	30 to 100 mg
Carrier frequency	50 to 100 MHz
Local mode pair	Two orthogonal shear modes
Local phase difference	+90 or -90 deg
Intermodule carrier phases	0, 120, 240 deg
Mechanical circularity	$C_{pol} \geq 0.95$
Chirality switching	1 to 10 Hz
Orientation	n perpendicular to g
Force direction	n x g
Mounting	Common nodal or symmetry-center support

The proposed engineering package uses three 12 mm by 12 mm resonator modules on a 60 mm equilateral triangle, a 10 mm central flexure, and a 90 mm outer package envelope. These dimensions are an engineering continuation intended to provide clear separation between modules, symmetric cable routing, and a compact force-balance interface. Final dimensions must be adjusted by finite-element modal analysis so the selected support is close to the relevant nodal manifold.

Proposed three-module HBAR force package - engineering dimensions

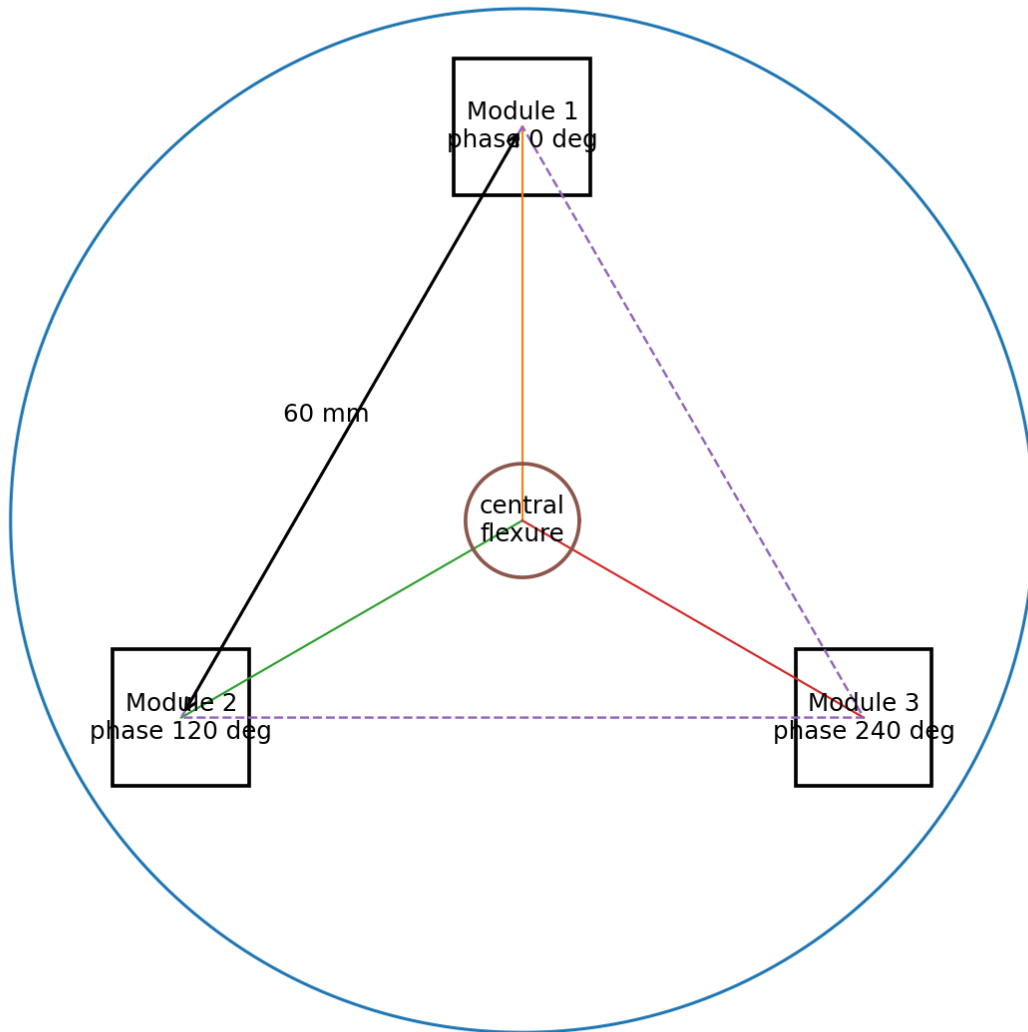


Figure 11. Dimensioned package proposal.

Why three modules are preferred

Two antipodal modules cancel the ordinary first-harmonic translation while their calculated lateral terms add. Three modules add a second cancellation: the doubled phase set also sums to zero, removing the ideal $2f$ ripple. When module positions are triangular and carrier phase is coordinated with position, the leading spatial moment and support torque also cancel.

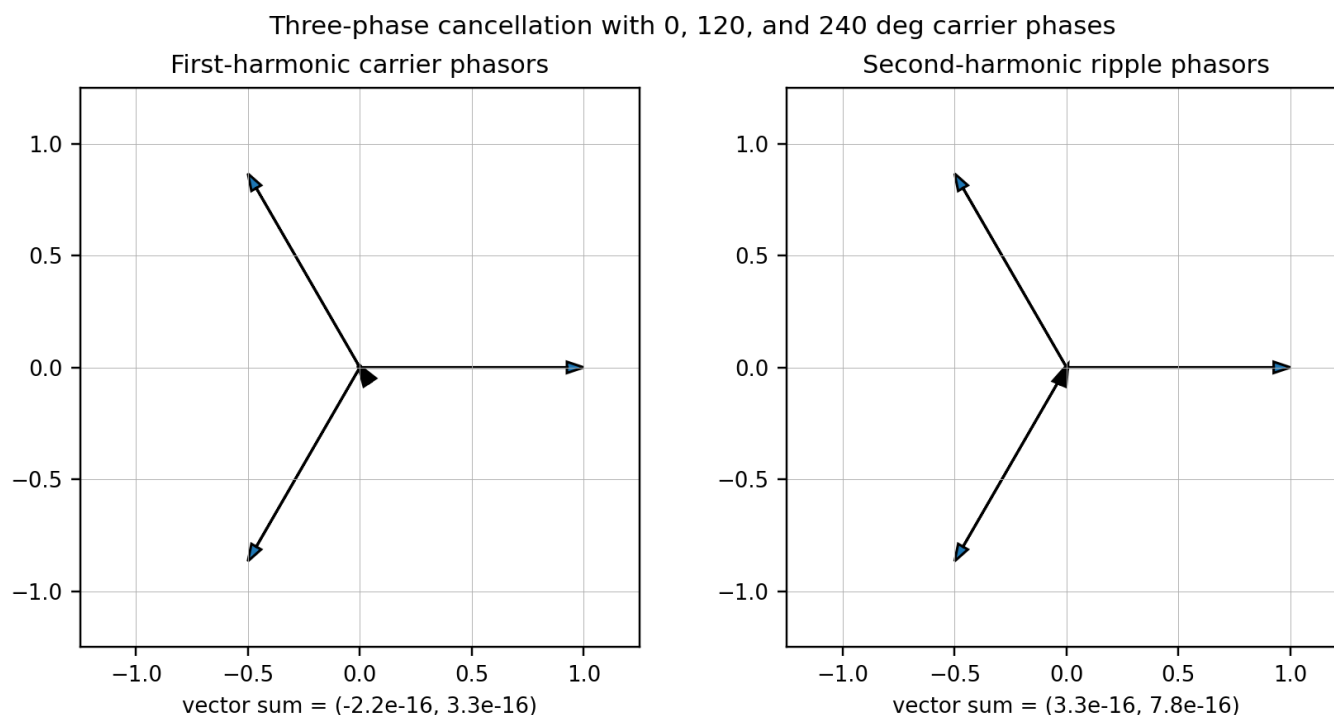


Figure 12. Exact first- and second-harmonic phasor cancellation.

The phase code changes temporal origin but not local chirality. Every module must retain the same sign of the local displacement-loop area. Reversing the quadrature sign in all modules reverses the complete lateral output without changing the three-phase cancellation.

8. Quantitative Output Envelope

Case	Active mass	Frequency	Ideal force	Force at C_pol=0.95, H=1e-3
Source tube, mode 5	2.261947e-02 kg	7.606366e+04 Hz	1.546638e-01 N	1.469306e-04 N
HBAR low case	3.000000e-05 kg	5.000000e+07 Hz	1.348405e-01 N	1.280985e-04 N
HBAR compact	1.000000e-05 kg	1.000000e+08 Hz	8.989366e-02 N	8.539898e-05 N
HBAR preferred	1.000000e-04 kg	1.000000e+08 Hz	8.989366e-01 N	8.539898e-04 N
MEMS/HBAR array	1.000000e-06 kg	1.000000e+09 Hz	8.989366e-02 N	8.539898e-05 N
SrTiO ₃ , 1 ng	1.000000e-12 kg	3.000000e+12 Hz	2.696810e-04 N	2.561969e-07 N
SrTiO ₃ , 1 microgram	1.000000e-09 kg	3.000000e+12 Hz	2.696810e-01 N	2.561969e-04 N

The preferred 100 mg, 100 MHz operating point has an ideal force of 0.8989366 N. With C_{pol} = 0.95, maximum orientation, and H = 1e-3, the calculated output is 0.85399 mN. With H = 1e-5, the output is 8.5399 microN. The lower 30 mg, 50 MHz point gives an ideal 0.13484 N and provides a less demanding first HBAR target.

The source fifth-mode whirling tube uses approximately 22.619 g of active steel and produces an ideal 154.66 mN. Three source-mass tubes would produce 463.99 mN using approximately 67.86 g active mass. A 100 mg, 100 MHz array exceeds that ideal output with about one thousandth of the active mass, provided the measured coherence factor remains sufficiently high.

9. Resonance Selection and Active-Mass Determination

The final operating overtone must not be selected from frequency alone. For each candidate mode j, measure or calculate its active mass, carrier frequency, circularity, and normalized force. Define the measured design product:

$$\Phi_{i,j} = M_{\text{active},j} f_j C_{\text{pol},j} H_j$$

$$H_j = F_j / [K_0 M_{\text{active},j} f_j C_{\text{pol},j} \sin(\beta)]$$

The optimum mode is the one maximizing Φ_j . A higher overtone should be rejected when reduced modal participation, localization, polarization impurity, or coherence loss outweighs the frequency increase. The active mass should be obtained from the measured mode shape or a calibrated finite-element model, not from total chip mass alone.

A practical mass estimate uses the measured complex displacement field $u(x)$. Normalize the field to its maximum or to stored modal energy and integrate the mass density with the chosen participation convention. The same convention must be used across the overtone ladder so H is comparable.

10. Amplitude Optimization and Mapping the Transfer Function H

Under the exact local circular law, displacement amplitude does not multiply the force. Ordinary inertial reaction, stored elastic energy, and dissipative power do depend strongly on amplitude. The optimal amplitude is therefore the smallest value that remains on the measured force plateau. Below that point, a coherence radius, localization threshold, finite response time, or other transfer process can reduce H .

At each frequency, sweep amplitude over as many decades as the metrology permits. Measure both displacement components, compute C_{pol} , and normalize the force by $M_{\text{active}} f C_{\text{pol}} \sin(\beta)$. A plateau identifies an amplitude-independent regime. A smooth roll-off locates a coherence transition. Oscillatory or stepped behavior would identify discrete localization or interference structure.

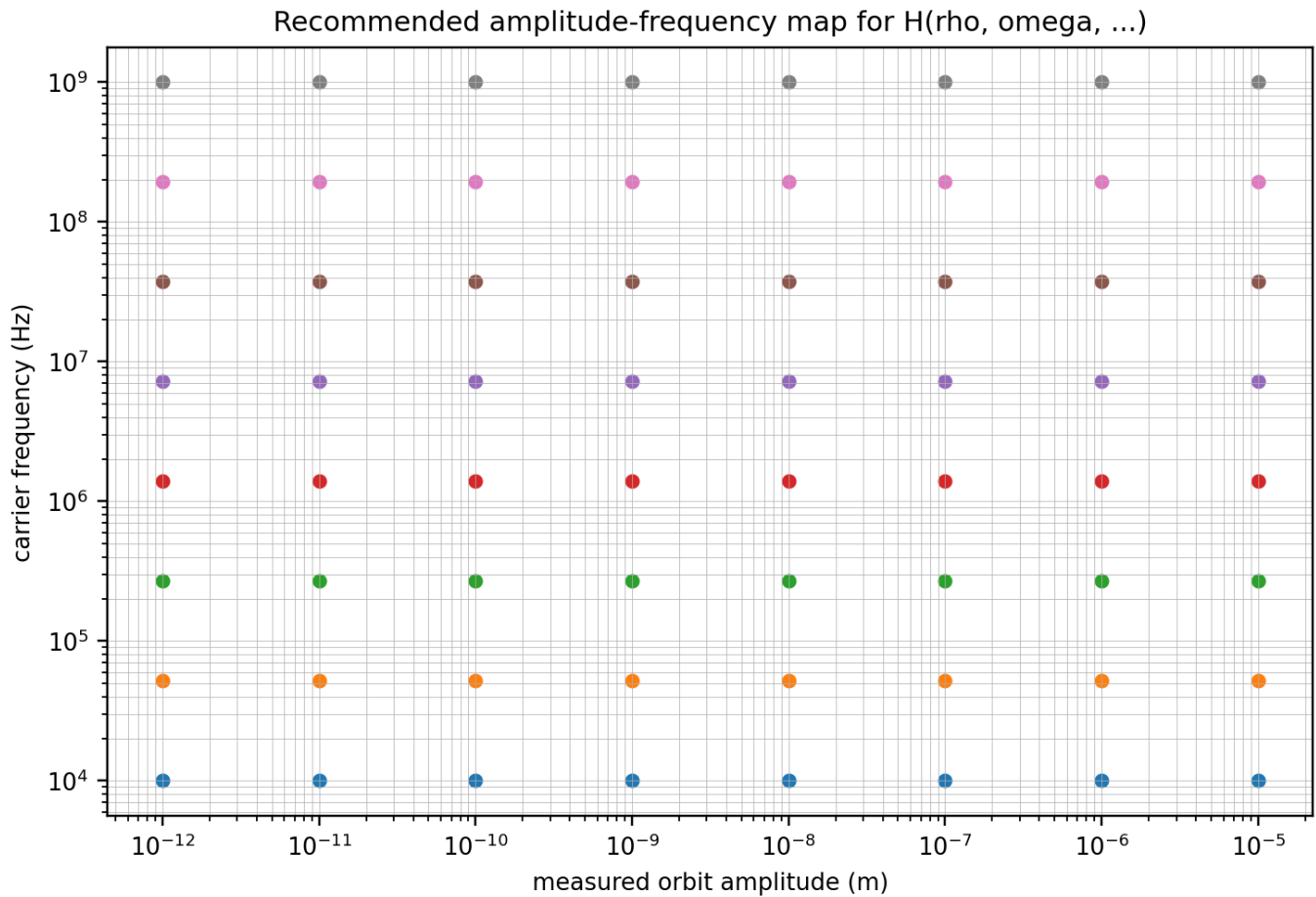


Figure 13. Recommended logarithmic amplitude-frequency sampling grid.

A complete H map should add wave vector, material, temperature, and boundary state as controlled axes. The overtone ladder changes frequency and spatial wavelength within one device. A second material at overlapping frequency separates material dependence from frequency dependence. Changes in mounting or boundary load test boundary sensitivity without changing the

active crystal.

11. Mechanical and Electrical Control System

Each module requires two independent actuator ports and at least two independent displacement observables. The electrical-to-mechanical response is a complex 2 by 2 matrix that varies with frequency and temperature. The controller identifies this matrix around the selected resonance pair and solves for the drive vector that produces the target physical circle.

$$\begin{aligned} u_{\text{measured}}(\omega) &= A_{\text{act}}(\omega) v_{\text{drive}} \\ v_{\text{drive}} &= A_{\text{act}}(\omega)^{-1} u_{\text{target}} \end{aligned}$$

The target is defined in measured displacement, not in voltage. A pair of phase-locked loops or a complex modal-state estimator holds equal amplitudes and the required quadrature. Common phase may be shifted without changing chirality. The second and third modules receive additional common carrier phases of 120 and 240 deg.

Cable forces must be treated as external mechanical channels. Use symmetric return paths, matched cable lengths, flexible low-force loops, and common-mode routing. Place current monitors close to the feedthrough and include dummy electrical loads so phase-switching transients can be distinguished from mechanical signals.

12. Chirality Modulation and Lock-In Readout

The desired mean is translated from nominal DC to a low modulation frequency by switching the relative mechanical phase from +90 to -90 deg. Carrier frequency, RMS displacement, and average drive power are held as constant as possible. Under the supplied law, the complete force vector reverses with chirality.

Chirality switching: +90 deg to -90 deg mechanical quadrature

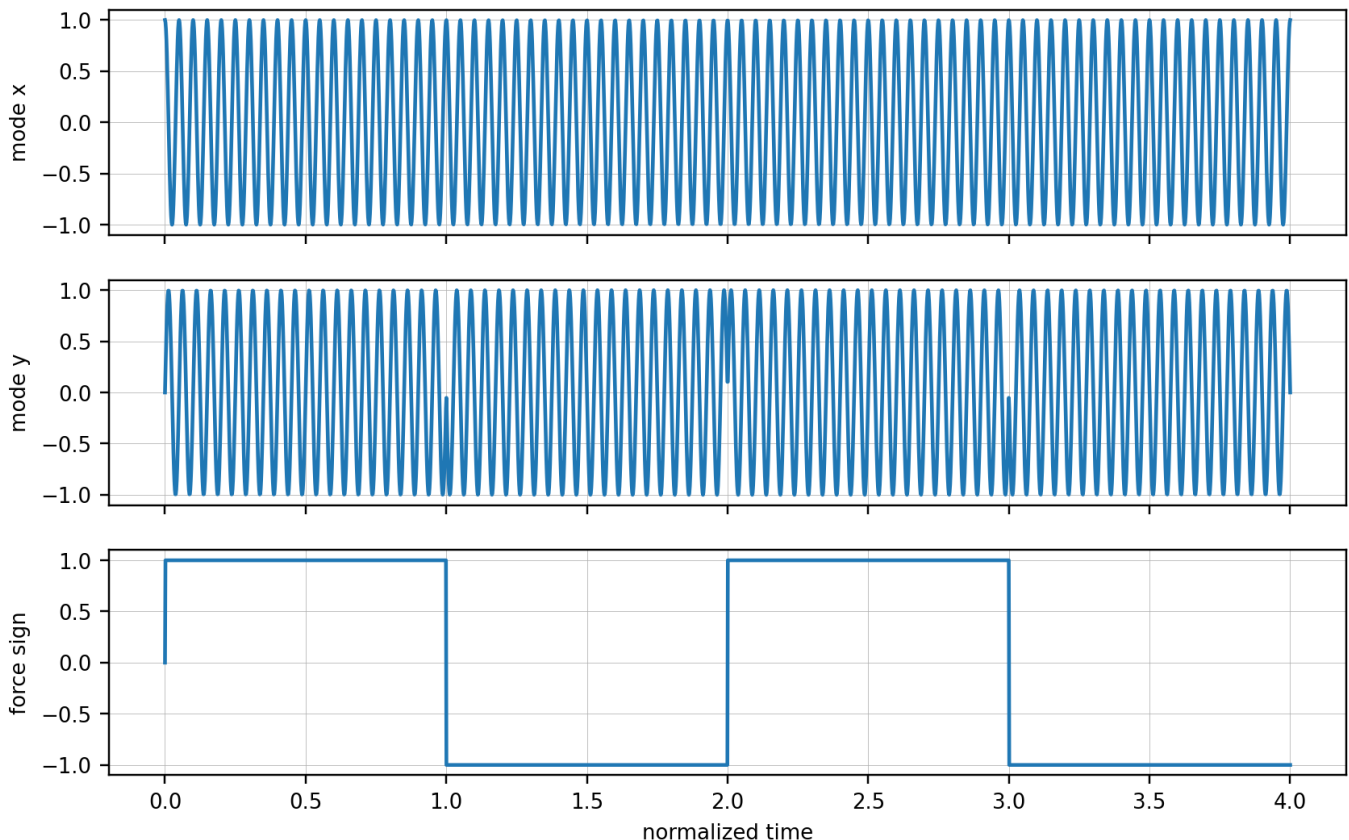


Figure 14. Chirality switching and force-sign lock-in reference.

The lock-in reference must be the measured sign and magnitude of C_{pol} , not the command bit. This prevents mode splitting, thermal drift, or amplifier phase error from creating a false correspondence between the nominal state and the actual mechanical state. Record in-phase and quadrature force channels at the switching frequency and at its harmonics.

13. Orientation and Vector Prediction

The vector law is stronger than a scalar force claim. The force magnitude follows $\sin(\beta)$, where β is the angle between the measured loop normal and g . The direction is $n \times g$. The source maximum-output normal is $n_{opt} = (-0.315865036407, 0.646162683024, -0.694768354089)$, with ICRS right ascension 116.050867377 deg and declination -44.008760899 deg. The corresponding lateral unit vector is (0.077469585422, 0.747375863362, 0.659869519070).

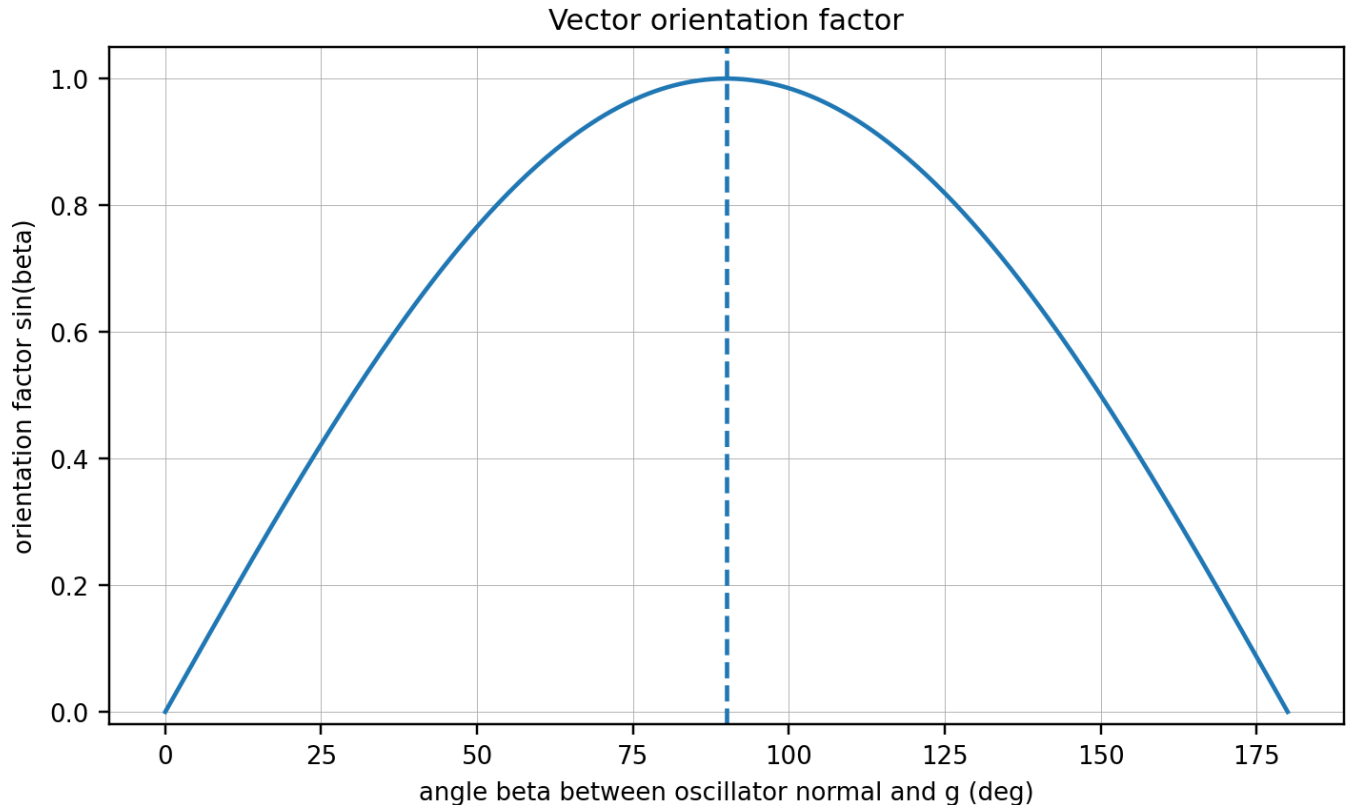


Figure 15. Predicted orientation factor.

The laboratory controller must transform the ICRS vectors into the local device frame using recorded UTC, site coordinates, and stage angles. Three configurations are decisive: n perpendicular to g for maximum output, n parallel to g for a vector null, and either $n \rightarrow -n$ or chirality reversal for sign reversal. A two-axis orientation sweep should reproduce both the changing magnitude and the rotating force direction.

14. Force Balance and Mechanical Metrology

The carrier frequency should lie far above the force-balance bandwidth. The force balance measures the slowly modulated reaction while independent sensors record carrier-frequency support acceleration and acoustic leakage. A three-axis balance is preferred because the two orthogonal null channels reveal vector misalignment and background coupling.

The minimum measurement set contains: two displacement components per module; carrier and harmonic support acceleration; triaxial force; voltage and current on every drive port; temperature at every module; package temperature gradients; acoustic emission; local magnetic field where relevant; stage orientation; UTC; and environmental pressure. Every force datum is accompanied by measured C_{pol} , active mass estimate, frequency, and orientation factor.

Subsystem	Required observable	Purpose
Two-axis displacement readout	Complex amplitude and phase of both shear modes	Compute C_{pol} and loop normal
Triaxial force balance	Force vector at chirality modulation frequency	Test magnitude and $n \times g$ direction
Package accelerometers	Carrier and harmonic reaction	Quantify residual mechanical imbalance
Electrical metrology	Per-port voltage, current, phase, power	Reject electrical transients and heating
Thermal sensing	Module and package temperature	Reject thermal gradients and track detuning
Orientation stage	Two angles plus absolute reference	Map $\sin(\beta)$ and vector rotation
Timing record	UTC synchronized acquisition	Transform field direction into lab frame

15. Decisive Controls and Background Rejection

Control transformation	Calculated K-Field outcome	Background discrimination
Relative phase +90 -> -90 deg	Complete force vector reverses	Preserves nominal carrier energy while changing chirality
Add 180 deg to both local axes	Mean unchanged	Changes carrier temporal phase but not chirality
Relative phase -> 0 or 180 deg	Mean tends to zero	Removes oriented area at comparable RMS motion
n parallel g	Vector null	Independent of frequency and carrier power
Rotate n over sphere	Force follows $n \times g$ and $\sin(\beta)$	Tests complete vector law
Reverse traveling-wave direction	Force reverses	Separates chirality from standing energy
Change material at matched kinematics	Mass law unchanged unless H is material dependent	Maps material universality
Frequency sweep at fixed mode family	Normalized force proportional to H	Maps high-frequency transfer
Amplitude sweep at fixed mode	Ideal plateau until H changes	Locates small-radius transition

Additional controls include dummy resonators with matched electrical impedance but no active mechanical circularity, reversal of common electrical polarity while preserving measured motion, cable re-routing tests, evacuated and atmospheric operation, mass-loading tests, and support-stiffness changes. A valid signal must remain tied to measured chirality and orientation rather than to a particular electrical wiring state.

16. Experimental Sequence

Phase	Procedure
Phase 0 - static metrology	Measure force-balance noise, thermal drift, cable force, magnetic sensitivity, optical momentum sensitivity, and stage-angle repeatability with all resonators inactive.
Phase 1 - single-module modal identification	Map the two orthogonal shear modes, frequency splitting, damping, actuator matrix, active mass, displacement field, and support reaction.
Phase 2 - single-module polarization sweep	Sweep relative phase from -180 to +180 deg at fixed RMS motion and compare force with measured C_{pol} .
Phase 3 - vector orientation sweep	Measure maximum, null, and sign-reversed orientations; then execute a two-axis grid.
Phase 4 - three-module cancellation	Operate 0, 120, and 240 deg carrier phases; trim complex gains to minimize carrier force and torque while retaining common chirality.
Phase 5 - overtone ladder	Repeat the complete normalized measurement over multiple overtones to determine $H(\omega, k)$.
Phase 6 - amplitude map	Sweep orbit amplitude downward until normalized force leaves the plateau.
Phase 7 - material and boundary replication	Repeat overlapping modes in a second resonator material and under altered boundary loading.
Phase 8 - terahertz extension	Apply counterpropagating circular THz excitation to SrTiO ₃ and compare helicity-odd force with the megahertz H map.

The measurement analysis follows derive, normalize, inspect residuals, correct measured kinematics, and repeat. No architecture-specific force coefficient is introduced. The smallest common transfer surface H that explains all devices is preferred over separate free parameters for each resonator.

17. Terahertz Material Extensions

The material ledger ranks SrTiO₃ first as a solid-state ionic-orbit platform. The reported source values include a 3 THz circular infrared-active soft-mode drive, approximately 0.5 THz drive bandwidth, ionic displacements of about 3 pm for oxygen and 1 pm for titanium/strontium, and a pump field of 230 kV/cm. At 3 THz, the ideal acceleration coefficient is 2.69681e8 m/s² per unit H, circularity, and orientation.

Coherent active mass at 3 THz	Ideal force
1 ng	0.269681 mN
1 microgram	0.269681 N

SrTiO₃ is the strongest small-mass endpoint but not the preferred first force apparatus. Picometre amplitude, finite optical penetration, pulsed THz momentum, thermal gradients, and the unknown high-frequency value of H make absolute force interpretation more difficult. The preferred experiment uses two back-to-back samples illuminated by equal counterpropagating beams. Electromagnetic momentum is arranged to cancel as a common-mode channel while phonon chirality is reversed.

LiNbO₃ provides reported chiral phonon frequencies near 6.770, 18.377, and 25.389 THz with reversible handedness. It is a higher-frequency material cross-check after the SrTiO₃ platform. YIG, CH₃F, optical-centrifuge O₂/N₂, and ultracold RbCs occupy complementary collective-spin, molecular-rotation, and long-coherence regimes but do not provide the shortest path to a compact room-scale force device.

Priority	Candidate	Frequency or coherence datum	Role
1	SrTiO ₃	3 THz driven circular soft mode	Solid-state ionic orbit
2	YIG	2 GHz normal-mode splitting; cooperativity 1e5	Collective spin angular momentum
3	CH ₃ F	51.072 GHz J=0 to 1 rigid-rotor transition	Ordinary polar molecular rotor
4	O ₂ /N ₂ optical centrifuge	10 THz final molecular rotation	Highest ordinary-gas rotation rate
5	LiNbO ₃	6.770, 18.377, 25.389 THz chiral modes	Reversible chiral phonons
6	Ultracold RbCs	0.78 s Ramsey coherence; >1.4 s spin echo	Long-coherence molecular reference

18. Laboratory Build Specification

A first build should be modular so that the force balance, resonator modules, orientation stage, and controller can be upgraded independently. The mechanical package must preserve symmetry and permit direct optical or capacitive access to both shear components. The electrical system requires six independently adjustable complex drive channels, although a shared frequency reference and common digital synthesis clock should be used.

Subsystem	Minimum specification
Resonator modules	Three matched dual-polarization bulk acoustic resonators with measurable orthogonal shear modes
Drive electronics	Six coherent RF channels with independent complex gain, common clock, and low-frequency chirality switching
Modal readout	Two displacement components per module; scanning laser vibrometry or equivalent complex metrology
Force balance	Three-axis flexure or electromagnetic balance with sensitivity spanning microN to mN at 1 to 10 Hz
Orientation stage	Two-axis stage capable of maximum, null, and full angular sweeps
Environmental enclosure	Thermal shielding, pressure control or vacuum capability, acoustic isolation, magnetic monitoring
Data acquisition	Synchronous recording of force, displacement, acceleration, voltage, current, temperature, orientation, and UTC
Control processor	Real-time estimation of C _{pol} , loop normal, carrier phase, and cancellation residuals

The first acceptance criterion is mechanical rather than force based: C_{pol} ≥ 0.95 in every module, carrier force and torque reduced by phase trim, and stable chirality switching without measurable change in RMS motion or average carrier power. Only after this state is demonstrated should the lateral force channel be interpreted.

19. Data Model and Normalized Analysis

Every measurement record should include raw and derived channels. Raw channels preserve the complete time series. Derived channels contain carrier frequency, two complex modal amplitudes per module, C_{pol} , loop normal, active mass estimate, orientation factor, chirality state, force vector, carrier reaction residuals, temperatures, and electrical power. The normalized observable is:

$$R = F_{\text{measured}} / [M_{\text{active}} f C_{\text{pol}} |n \times g|]$$

Under the local source law and $H = 1$, R equals $\pi \kappa_F$. More generally, the dimensionless transfer factor is $H = R / (\pi \kappa_F)$. Residual analysis should plot H against amplitude, frequency, wave vector, modal order, material, temperature, boundary condition, and package-reaction metrics.

A common collapse of normalized data across the whirling tube, HBAR overtones, annulus, MEMS arrays, and terahertz materials identifies a scale-invariant regime. A systematic transition identifies the coherence or localization boundary. Mode-dependent structure correlated with reciprocal-gradient magnitude would distinguish a nonlocal Fourier-gradient mechanism from a purely local kinematic kernel.

20. Findings and New Discovery Pathways

The combined analysis changes the device-design problem from maximizing rigid-rotor speed to maximizing coherent local chirality. The relevant state is a closed oriented displacement loop, not continuous material circulation. This removes bearings, shafts, and rotor balance from the minimum mechanism and opens a direct route to elastic waves, bulk acoustic overtones, MEMS arrays, and chiral phonons.

The primary hidden design law is that temporal phase can cancel ordinary motion without canceling the local chiral mean. The three-phase architecture is therefore not merely a vibration reduction technique; it is the mechanism that separates the predicted phase-independent lateral channel from first- and second-harmonic package reaction.

The second hidden design law is that minimal total mass and maximal force are not conflicting when the distinction between active and inactive mass is maintained. The optimized device has a high active-mass fraction and a light package. Milligram-scale HBAR mass at megahertz frequency can outperform gram-scale kilohertz structures in the ideal law.

The third discovery pathway is the direct experimental reconstruction of H . Rather than assuming scale invariance to the atomic limit, the program maps the transition. The same normalized observable connects macroscopic beams, high-overtone substrates, surface waves, MEMS arrays, and terahertz ionic orbits. This creates a single quantitative framework for identifying coherence length, frequency response, material dependence, and nonlocal gradient structure.

21. Conclusions

The optimal laboratory-scale approach is a three-module dual-shear high-overtone bulk acoustic resonator array. Each module executes circularly polarized elastic motion from two orthogonal shear modes in quadrature. The modules use carrier phases of 0, 120, and 240 deg while retaining one common chirality. This cancels ordinary carrier reaction, leading torque, and the ideal 2f ripple while preserving the phase-independent lateral mean.

The governing engineering objective is $\Phi = f \eta_m C_{pol} H \sin(\beta)$. Active resonator mass should be preserved; inactive mass should be minimized. A preferred 30 to 100 mg active mass at 50 to 100 MHz gives an ideal force range from 0.135 to 0.899 N. The actual output is obtained by multiplying by measured circularity, orientation, and H .

The first experimental priority is to establish a mechanically calibrated circular state, measure chirality reversal and the vector orientation law, and suppress carrier reaction through three-phase balance. The second priority is to sweep overtones and amplitudes to map H . The third priority is replication across materials and boundary states. SrTiO₃ and LiNbO₃ then extend the same displacement-loop primitive into the terahertz ionic and chiral-phonon regime.

The complete development path is therefore: three-phase dual-shear HBAR, overtone and amplitude optimization, cross-material replication, and terahertz coherent-ion scale testing. This path maximizes laboratory force output while simultaneously generating the measurements required to reconstruct the underlying coherence and localization structure.

Appendix A. Authoritative Numerical Input Ledger

Quantity	Value
Source schema 1	kfield.circularly_polarized_elastic_oscillator.v1
Source schema 2	kfield.materials.coherent_angular_momentum.v1
K0	8.989366088879062e-05 N/(kg Hz)
kappa_F	3.0740508997020345e-11
g	9.308252172704e+05 m/s
Source tube mass	0.022619467105847 kg
Source tube frequency	833.333333333333 Hz
Source tube force	0.001694455587915 N
Cell volume	0.117914891916699 nm ³
Metric equivalence max error	1.14023235e-11

Appendix B. Complete 27-Node ICRS Cell3 Ledger

Address	lambda	Polarity	Kind	x (nm)	y (nm)	z (nm)
(-1,-1,-1)	-3	-1	corner	-0.176351350144	0.118682429904	0.637311498326
(-1,-1,0)	-2	1	edge center	-0.181416008957	-0.076211421278	0.292444294144
(-1,0,-1)	-2	1	edge center	-0.158685962221	0.222342355125	0.36361095563
(0,-1,-1)	-2	1	edge center	-0.012600729109	0.091233925961	0.618567746877
(-1,-1,1)	-1	-1	corner	-0.18648066777	-0.271105272459	-0.052422910037
(-1,0,0)	-1	-1	face center	-0.163750621035	0.027448503943	0.018743751448
(-1,1,-1)	-1	-1	corner	-0.141020574299	0.326002280346	0.089910412934
(0,-1,0)	-1	-1	face center	-0.017665387922	-0.103659925221	0.273700542696
(0,0,-1)	-1	-1	face center	0.005064658813	0.194893851182	0.344867204181
(1,-1,-1)	-1	-1	corner	0.151149891926	0.063785422018	0.599823995429
(-1,0,1)	0	1	edge center	-0.168815279848	-0.167445347239	-0.326123452733
(-1,1,0)	0	1	edge center	-0.146085233113	0.131108429164	-0.254956791248
(0,-1,1)	0	1	edge center	-0.022730046735	-0.298553776403	-0.071166661485
(0,0,0)	0	1	body center	0	0	0
(0,1,-1)	0	1	edge center	0.022730046735	0.298553776403	0.071166661485
(1,-1,0)	0	1	edge center	0.146085233113	-0.131108429164	0.254956791248
(1,0,-1)	0	1	edge center	0.168815279848	0.167445347239	0.326123452733
(-1,1,1)	1	-1	corner	-0.151149891926	-0.063785422018	-0.599823995429
(0,0,1)	1	-1	face center	-0.005064658813	-0.194893851182	-0.344867204181
(0,1,0)	1	-1	face center	0.017665387922	0.103659925221	-0.273700542696
(1,-1,1)	1	-1	corner	0.141020574299	-0.326002280346	-0.089910412934
(1,0,0)	1	-1	face center	0.163750621035	-0.027448503943	-0.018743751448
(1,1,-1)	1	-1	corner	0.18648066777	0.271105272459	0.052422910037
(0,1,1)	2	1	edge center	0.012600729109	-0.091233925961	-0.618567746877
(1,0,1)	2	1	edge center	0.158685962221	-0.222342355125	-0.36361095563
(1,1,0)	2	1	edge center	0.181416008957	0.076211421278	-0.292444294144
(1,1,1)	3	-1	corner	0.176351350144	-0.118682429904	-0.637311498326

Appendix C. Complete Coordinate-Plane Registry

Channel a; opposite face family bc

Property	Value
Unit normal	-0.995965190309, 0.083602162507, -0.032619290564
Primitive spacing	0.329546529709293 nm
Centered spacing	0.164773264854647 nm
Face area	0.357809539128562 nm ²

Plane	Nine addresses
-1	(-1,-1,-1); (-1,-1,0); (-1,-1,1); (-1,0,-1); (-1,0,0); (-1,0,1); (-1,1,-1); (-1,1,0); (-1,1,1)
0	(0,-1,-1); (0,-1,0); (0,-1,1); (0,0,-1); (0,0,0); (0,0,1); (0,1,-1); (0,1,0); (0,1,1)
1	(1,-1,-1); (1,-1,0); (1,-1,1); (1,0,-1); (1,0,0); (1,0,1); (1,1,-1); (1,1,0); (1,1,1)

Channel b; opposite face family ac

Property	Value
Unit normal	0.089052594102, 0.866576769291, -0.491033948326
Primitive spacing	0.45159737972413 nm
Centered spacing	0.225798689862065 nm
Face area	0.261106235799531 nm ²

Plane	Nine addresses
-1	(-1,-1,-1); (-1,-1,0); (-1,-1,1); (0,-1,-1); (0,-1,0); (0,-1,1); (1,-1,-1); (1,-1,0); (1,-1,1)
0	(-1,0,-1); (-1,0,0); (-1,0,1); (0,0,-1); (0,0,0); (0,0,1); (1,0,-1); (1,0,0); (1,0,1)
1	(-1,1,-1); (-1,1,0); (-1,1,1); (0,1,-1); (0,1,0); (0,1,1); (1,1,-1); (1,1,0); (1,1,1)

Channel c; opposite face family ab

Property	Value
Unit normal	0.194092200808, 0.913177168821, 0.358379234791
Primitive spacing	0.605097741514743 nm
Centered spacing	0.302548870757372 nm
Face area	0.1948691654699 nm ²

Plane	Nine addresses
-1	(-1,-1,-1); (-1,0,-1); (-1,1,-1); (0,-1,-1); (0,0,-1); (0,1,-1); (1,-1,-1); (1,0,-1); (1,1,-1)
0	(-1,-1,0); (-1,0,0); (-1,1,0); (0,-1,0); (0,0,0); (0,1,0); (1,-1,0); (1,0,0); (1,1,0)
1	(-1,-1,1); (-1,0,1); (-1,1,1); (0,-1,1); (0,0,1); (0,1,1); (1,-1,1); (1,0,1); (1,1,1)

Appendix D. Seven-Sheet Lambda Ledger

lambda	Count	Addresses
-3	1	(-1,-1,-1)
-2	3	(-1,-1,0); (-1,0,-1); (0,-1,-1)
-1	6	(-1,-1,1); (-1,0,0); (-1,1,-1); (0,-1,0); (0,0,-1); (1,-1,-1)
0	7	(-1,0,1); (-1,1,0); (0,-1,1); (0,0,0); (0,1,-1); (1,-1,0); (1,0,-1)
1	6	(-1,1,1); (0,0,1); (0,1,0); (1,-1,1); (1,0,0); (1,1,-1)
2	3	(0,1,1); (1,0,1); (1,1,0)
3	1	(1,1,1)

Appendix E. Compact Formula Index

Name	Formula
Centered node	$p_L = B L / 2$
Physical response tensor	$G_{\text{phase}} = 4 B^{(-T)} W B^{(-1)}$
Field response	$g = G_{\text{phase}} V_E$
Cycle increment	$\Delta v_K = \pi \kappa_F \sigma (n \times g)$
Mean acceleration	$\langle a_K \rangle = \pi \kappa_F f \sigma (n \times g)$
Mechanical circularity	$C_{\text{pol}} = 2 A B \sin(\phi) / (A^2 + B^2)$
General force	$F_K = K_0 M_{\text{active}} f C_{\text{pol}} H \sin(\beta)$
Force coefficient	$K_0 = \pi \kappa_F g $
Active mass fraction	$\eta_m = M_{\text{active}} / M_{\text{total}}$
Specific objective	$F_K / M_{\text{total}} = K_0 f \eta_m C_{\text{pol}} H \sin(\beta)$
Measured transfer	$H = F / [K_0 M_{\text{active}} f C_{\text{pol}} \sin(\beta)]$
Normalized response	$R = F / [M_{\text{active}} f C_{\text{pol}} n \times g]$

Appendix F. Complete Calculated Force Cases

Case	Mass (kg)	Frequency (Hz)	M f (kg Hz)	Ideal force (N)
Source tube, mode 5	0.022619467	7.606365604e+04	1720.519365859	0.154663784
HBAR low case	3.000000000e-05	5.000000000e+07	1500	0.134840491
HBAR compact	1.000000000e-05	1.000000000e+08	1000	0.089893661
HBAR preferred	1.000000000e-04	1.000000000e+08	1.000000000e+04	0.898936609
MEMS/HBAR array	1.000000000e-06	1.000000000e+09	1000	0.089893661
SrTiO ₃ , 1 ng	1.000000000e-12	3.000000000e+12	3	2.696809827e-04
SrTiO ₃ , 1 microgram	1.000000000e-09	3.000000000e+12	3000	0.269680983

Appendix G. References

No.	Reference
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18	Krytronx K-Field historical document library: authoritative K-Field Base Cell matrix and Lucas node/plane conventions. Evidence basis: B_nm matrix, 27-node centered address set, reciprocal plane family, sheet spacing and polarity rule.

Appendix H. Archival Record

The companion machine-readable file records source hashes, numerical inputs, derived force cases, engineering proposal parameters, figure hashes, formulas, and the output PDF hash. The PDF uses ReportLab and only built-in core fonts for document text. Every table has explicit page-width-based column widths, Paragraph-wrapped cells, and repeating header rows.